Introduction to Algorithms & C Programming Fundamentals

Comprehensive Lab Notes: Algorithms, Data Types, Operators, Control Structures, and Functions

1. Algorithms and Flowcharts (Basics)

An algorithm is a step-by-step sequence of instructions designed to solve a specific problem. A flowchart is a visual representation of an algorithm using standardized symbols:

- **Oval:** Start / End points.
- **Paralellogram:** Input / Output operations.
- **Rectangle:** Processing steps and variable assignments.
- **Rhombus (Diamond):** Decision making and comparison (Yes/No branches).
- **Flowlines:** Direction of execution flow.

2. Data Types and Variables in C

Data types define the storage size and range of values a variable can hold. C provides fundamental integer, floating-point, character, and boolean types.

Data Type	Memory Size	Minimum Value	Maximum Value
`int`	4 bytes	-2,147,483,648	2,147,483,647
`unsigned int`	4 bytes	0	4,294,967,295
`float`	4 bytes	1.2E-38	3.4E+38
`double`	8 bytes	2.3E-308	1.7E+308
`char`	1 byte	-128	127

3. Operators in C

Operators perform operations on variables and values:

- **Arithmetic Operators:** +, -, *, /, %, and increment/decrement (`++`, `--`). Pre/post operators determine evaluation timing.
- **Assignment Operators:** +=, -=, *=, /=, %=.
- **Comparison Operators:** ==, !=, `>`, `<`, `>=`, `<=`. Return 1 (true) or 0 (false).
- **Logical Operators:** && (AND), || (OR), ! (NOT).
- **Bitwise Operators:** &, |, ^, ~, << (Left shift multiplies by 2), >> (Right shift divides by 2).

4. Functions in C

Functions are self-contained code blocks that perform specific tasks to promote code reuse and modularity.

```
<return_type> function_name(parameters) {
    // Function body
    return value; // Optional based on return type
}
```

Function Types: Void/No parameters, Void with parameters, Value-returning with no parameters, Value-returning with parameters.